WORKED SOLUTION

STEP II 2025, Question 7

Pure • Nonlinear Differential Equations

STEP II 2025, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

The right-hand side $2x\dfrac{\mathrm{d}x}{\mathrm{d}t}$ is exactly the derivative of x^2 with respect to t, so the equation integrates at once to a first-order equation $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2 + C$. The starting conditions fix C, and the three parts are the three possible signs of C: zero, positive and negative. Each first-order equation then separates, and the behaviour of the particle can be read off from the formula, or even straight from the sign of $x^2 + C$.

Solution

Integrating once

Since $\dfrac{\mathrm{d}}{\mathrm{d}t}\left(x^2\right) = 2x\dfrac{\mathrm{d}x}{\mathrm{d}t}$, the equation says

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(x^2\right), \qquad \text{so} \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = x^2 + C$$

for a constant C. At $t = 0$ we have $x = a$, so C is the initial velocity minus a^2. In the three parts $C = 0$, $C = p^2$ and $C = -q^2$ respectively.

Part (i)

Here $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2$. The velocity is positive whenever $x \neq 0$, so x increases from $a > 0$ and never reaches 0; we may therefore divide by x^2 and separate:

$$\int_a^x \frac{\mathrm{d}u}{u^2} = t \quad \Longrightarrow \quad \frac{1}{a} - \frac{1}{x} = t \quad \Longrightarrow \quad x = \frac{a}{1 - at}.$$

As t increases from 0 towards $\frac{1}{a}$, the particle moves in the positive direction ever faster (its velocity is x^2), and $x \to \infty$ as $t \to \frac{1}{a}$. The particle escapes to infinity in the finite time $\frac{1}{a}$, and the solution does not exist beyond that time.

Part (ii)

Now $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2 + p^2$, which is always positive, so we can separate:

$$\int_a^x \frac{\mathrm{d}u}{u^2+p^2} = t \quad \Longrightarrow \quad \frac{1}{p}\left(\arctan\frac{x}{p} - \arctan\frac{a}{p}\right) = t.$$

Writing $\alpha = \arctan\dfrac{a}{p}$, which lies in $(0, \frac{\pi}{2})$ because $a, p > 0$,

$$x = p\tan(pt+\alpha).$$

This holds while $pt + \alpha < \frac{\pi}{2}$. The particle again moves in the positive direction with increasing speed, and $x \to \infty$ as $pt + \alpha \to \frac{\pi}{2}$, that is, as

$$t \to \frac{1}{p}\left(\frac{\pi}{2} - \arctan\frac{a}{p}\right) = \frac{1}{p}\arctan\frac{p}{a},$$

using $\arctan\theta + \arctan\frac{1}{\theta} = \frac{\pi}{2}$ for $\theta > 0$. So it escapes to infinity in a finite time, sooner than in (i), as you'd expect from the larger initial velocity.

Part (iii)

Now

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x^2 - q^2 = (x-q)(x+q).$$

The velocity is zero at $x = \pm q$, so these two positions are equilibria. The sign of $x^2 - q^2$ already tells us the story: the particle moves right when $x > q$, left when $-q < x < q$. So the answer depends on how a compares with q.

Separate using partial fractions, $\dfrac{1}{x^2-q^2} = \dfrac{1}{2q}\left(\dfrac{1}{x-q} - \dfrac{1}{x+q}\right)$:

$$\frac{1}{2q}\ln\left|\frac{x-q}{x+q}\right| = t + \text{constant}.$$

Exponentiating and fixing the constant by $x = a$ at $t = 0$ suggests

$$\frac{x-q}{x+q} = Ke^{2qt}, \qquad K = \frac{a-q}{a+q}, \qquad \text{i.e.} \qquad x = q\,\frac{1+Ke^{2qt}}{1-Ke^{2qt}}.$$

To be sure the signs are right in every case, check this directly. Put $E = Ke^{2qt}$, so $\frac{\mathrm{d}E}{\mathrm{d}t} = 2qE$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} = q\cdot\frac{2\,\frac{\mathrm{d}E}{\mathrm{d}t}}{(1-E)^2} = \frac{4q^2E}{(1-E)^2}, \qquad x^2 - q^2 = q^2\,\frac{(1+E)^2-(1-E)^2}{(1-E)^2} = \frac{4q^2E}{(1-E)^2},$$

which agree. At $t = 0$, $1 + K = \frac{2a}{a+q}$ and $1 - K = \frac{2q}{a+q}$, so $x = q\cdot\frac{a}{q} = a$. So this formula is the solution, wherever $1 - E \neq 0$. Note that $-1 < K < 1$ since $a > 0$.

Case $a > q$. Then $0 < K < 1$ and $E = Ke^{2qt}$ increases from K. The particle starts with positive velocity $a^2 - q^2$ and moves in the positive direction, faster and faster. The denominator $1 - E$ reaches 0 when $e^{2qt} = \frac{1}{K}$, so

$$x \to \infty \quad \text{as} \quad t \to \frac{1}{2q} \ln \frac{a+q}{a-q}.$$

As in (i) and (ii), the particle escapes to infinity in finite time.

Case $a = q$. Then $K = 0$ and $x = q = a$ for all t. The initial velocity is 0, and so is the acceleration $2x\frac{\mathrm{d}x}{\mathrm{d}t}$: the particle stays at rest at $x = a$.

Case $0 < a < q$. Then $-1 < K < 0$, so $E < 0$ and the denominator $1 - E > 1$ never vanishes: the solution exists for all $t \geq 0$.

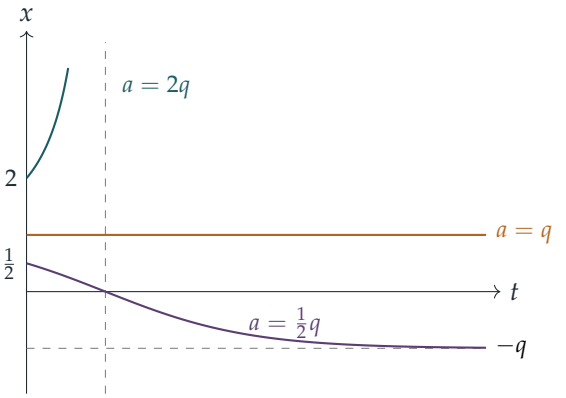
Ask yourself Does the particle stop at the origin, or go through it, and where does it end up?

The initial velocity $a^2 - q^2$ is negative, so the particle moves in the negative direction. It is at the origin when $E = -1$, i.e. at $t = \frac{1}{2q} \ln \frac{q+a}{q-a}$, and it passes straight through, since the velocity there is $-q^2 \neq 0$. As $t \to \infty$, $E \to -\infty$ and

$$x = q\,\frac{1+E}{1-E} \to -q.$$

Throughout, $|x| < q$ (because $\frac{1+E}{1-E} \in (-1, 1)$ for $E < 0$), so the velocity $x^2 - q^2$ stays negative: the particle moves steadily to the left and approaches $x = -q$ without ever reaching it. Its speed $q^2 - x^2$ is greatest, q^2, as it passes the origin, and then dies away to zero.

In hyperbolic form these solutions are $x = -q \coth\left(qt - \operatorname{arcoth} \frac{a}{q}\right)$ for $a > q$ and $x = -q \tanh\left(qt - \operatorname{artanh} \frac{a}{q}\right)$ for $a < q$. The sketch shows the three cases with $q = 1$.



For $a = 2q$ and $a = \frac{1}{2}q$ the special time $\frac{1}{2q} \ln 3$ (dashed vertical line) is the blow-up time and the time of crossing the origin respectively.

Checks

- Letting $p \to 0$ in (ii): $p\tan(pt + \arctan\frac{a}{p}) \to \frac{a}{1-at}$ and $\frac{1}{p}\arctan\frac{p}{a} \to \frac{1}{a}$, matching (i). The same happens as $q \to 0$ in (iii) with $a > q$.
- Differentiate $x = \frac{a}{1-at}$: $\dot{x} = \frac{a^2}{(1-at)^2} = x^2$ and $\ddot{x} = \frac{2a^3}{(1-at)^3} = 2x\dot{x}$.
- Stepping the original second-order equation forward numerically for, say, $a = 0.5$, $q = 1.4$ gives $x \approx -1.0814$ at $t = 1$ and $x \approx -1.3987$ at $t = 3$, agreeing with the formula, and the particle is at the origin at $t \approx 0.2669$, as predicted.

Common slips

- Not noticing that $2x\frac{\mathrm{d}x}{\mathrm{d}t}$ is an exact derivative, and trying a substitution such as $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ with $v\frac{\mathrm{d}v}{\mathrm{d}x}$, which works but takes longer.
- Dropping the modulus in the logarithm in (iii) and so getting the wrong sign of K when $a < q$.
- Missing the equilibrium case $a = q$, or saying the particle "stops at $-q$" when it only approaches it.

Beyond STEP

The finite-time blow-up seen here is a hallmark of nonlinear equations: $\dot{x} = x^2$ has a perfectly smooth right-hand side, yet its solutions only exist for a limited time, something that never happens for linear equations. Part (iii) is a first look at phase-line analysis: the equilibria $x = q$ (unstable) and $x = -q$ (stable) organise all the motion, and you can predict it from the sign of $x^2 - q^2$ before solving anything. Boyce and DiPrima's *Elementary Differential Equations* treats both ideas, including existence and uniqueness of solutions.

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